

BUDDY PLAYER FOR VISUALLY CHALLENGED

Bisquare develops a low-cost, handheld media player that can read out the books aloud



■ JALAJA RAMANUNNI

At a conference, the CEO of a product design company noticed nothing peculiar—attendees discussed issues and took notes on their laptops. It was only after the meeting that he realised that one of them was visually impaired. The CEO wondered how the man managed to use the laptop like others. He wanted to find out more and approached the man, who turned out to be the head of IT department of the National Association for the Blind (NAB), Dipendra Manocha.

Manocha exclaimed that there were many devices in the market for the visually impaired, but most of them were expensive and beyond reach. This inspired the CEO of Bisquare, Ramendra Baoni, to work out a low-cost product to make technology accessible to the visually impaired. As a result,

his company designed and developed Buddy player—a handheld media player that can be used by the visually impaired with ease. The media player is integrated with a digital talking book that can read out the menu and books aloud. The device is developed as a platform for education that allows the user to carry the learning material (e-books) on a media player.

What sets it apart

Buddy player has won the national award for 'product innovation for the physically challenged' from the ministry of social justice and empowerment. Priced at Rs 4000, its cost is one of its biggest USPs. Bisquare claims it to be the cheapest Daisy reader available for the visually impaired, with the closest competitor priced at about Rs 16,000.

"The price was deliberately kept low, knowing that most of the differ-

ently-abled cannot afford expensive devices. Conventionally, the device is designed using semiconductors or system-on-chip designed in the US or Europe, but as our price point was aggressive, we decided to consider other options," Baoni reveals.

How it was made low-cost

In an effort to maintain the low-price tag, Baoni went to China where he evaluated the market for low-cost MP3 players. He visited design houses that made low-cost devices and met the key people there.

Baoni points out, "I realised that most of their core chip and price points are really good but they lagged behind in the quality of the entire product, like peripherals and plastic."

He decided to adopt a highly integrated system-on-chip built by Actions Semiconductor and build a device around it. The chip includes a digital signal processor, audio codec, audio amplifier and power section. This reduced the number of components outside, which, in turn, helped to reduce the overall cost of the device. They also made sure not to over-design the product and fill it with unnecessary features.

Challenges in integrating Daisy reader

In the generic media player, Bisquare integrated the Daisy reader—a digital talking book technology that was developed in the US and Japan. Daisy is an international standard that has been adopted in India by the NAB, and now all the NCERT syllabus books from Class IX to XII are available in this format. Besides, a lot of books, magazines